

Title page

Project Title: Hotel Booking System (Java Swing + OOP)

Problem Statement:

Manual hotel booking systems are inefficient and error-prone. This project automates room booking, customer management, and data storage.

Target User:

- Hotel staff
- Managers
- Small hotel/lodge administrators

System Design

High-Level Architecture:

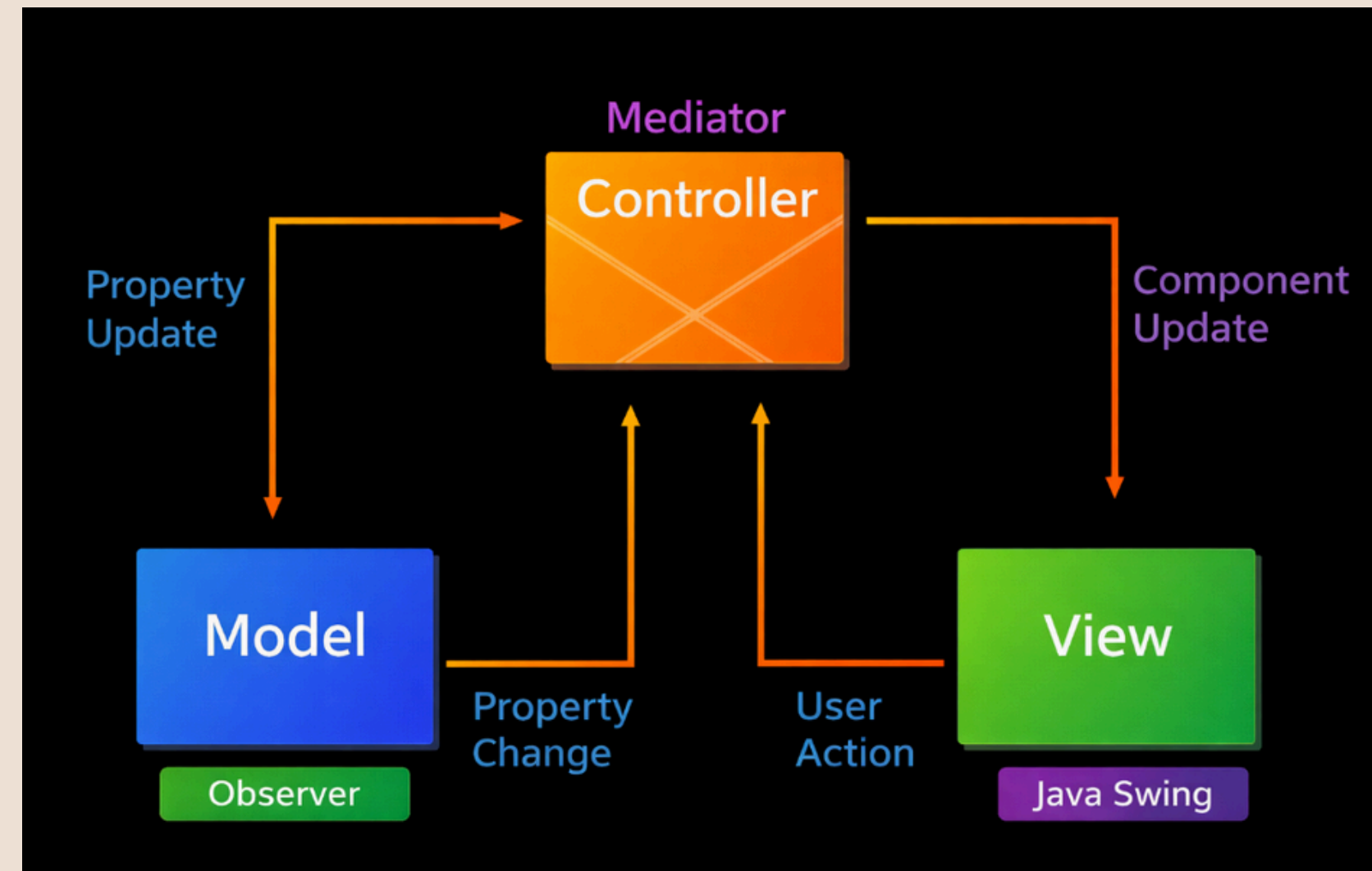
- Follows MVC structure
- Model → Room, Customer, Booking
- Service → HotelService, FileService
- View → HotelGUI

Key Classes & Responsibilities:

- Room → stores room details
- Customer → stores customer info
- Booking → links room + customer
- HotelService → core logic
- SaveThread → background saving

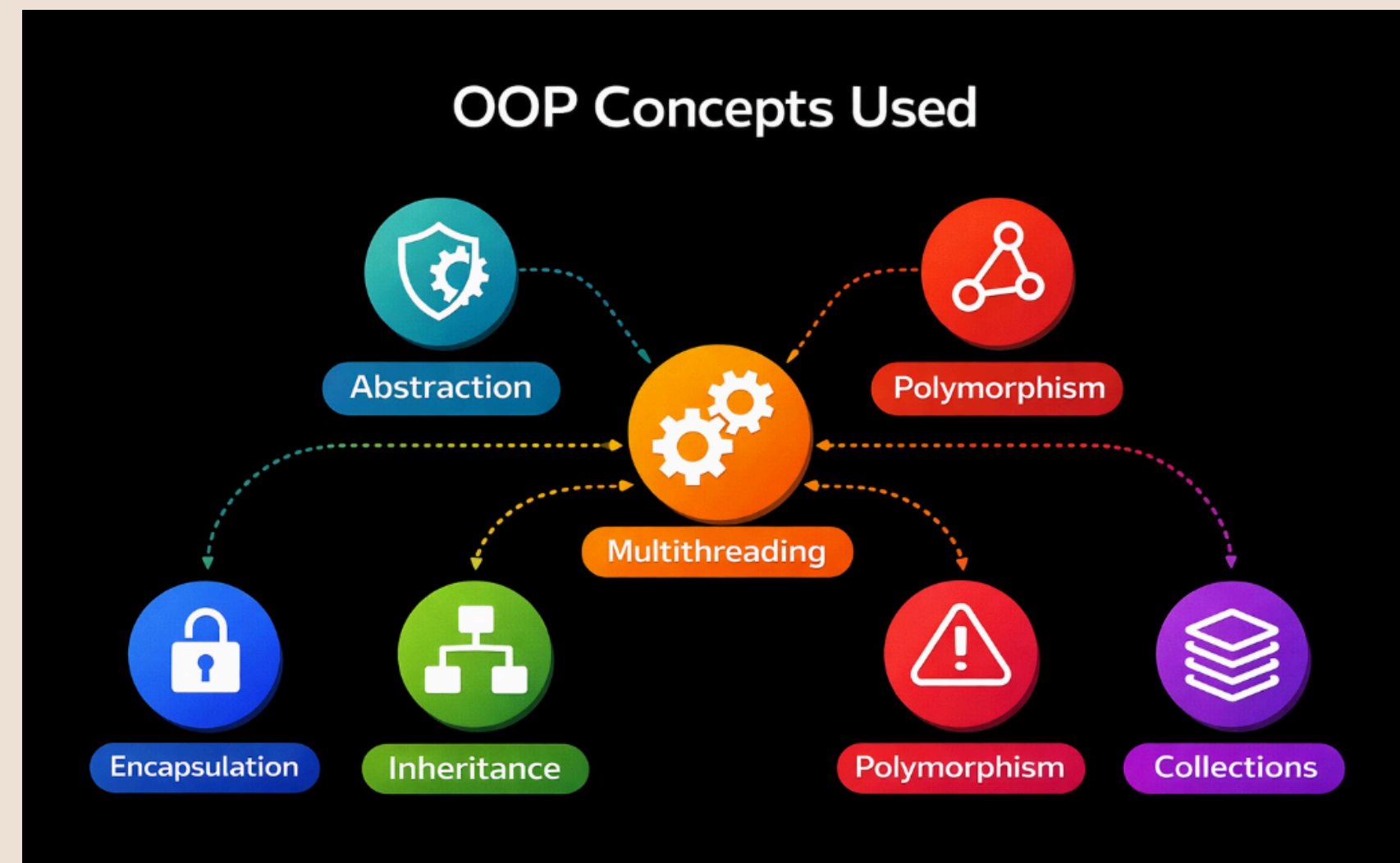
Package Structure:

- Main → entry point
- Model → data classes
- Service → business logic
- View → GUI



OOP Concepts Used

- **Abstraction:**
 - HotelService class
 - Methods like bookRoom(), addRoom() hide internal logic
- **Inheritance:**
 - SaveThread class
 - Extends Thread class
- **Polymorphism:**
 - display() method in Room class
 - Can be overridden for different room types
- **Exception handling:**
 - FileService class,
 - Used in file operations with try-catch
- **Multithreading:**
 - SaveThread class
 - Runs background saving process
- **Collections:**
 - ArrayList in HotelService
 - Stores rooms and bookings
- **Design Pattern:**
 - Follows MVC (Model-View-Controller)



Technical Highlight

Design Decision:

- Using multithreading for background saving

Code Logic:

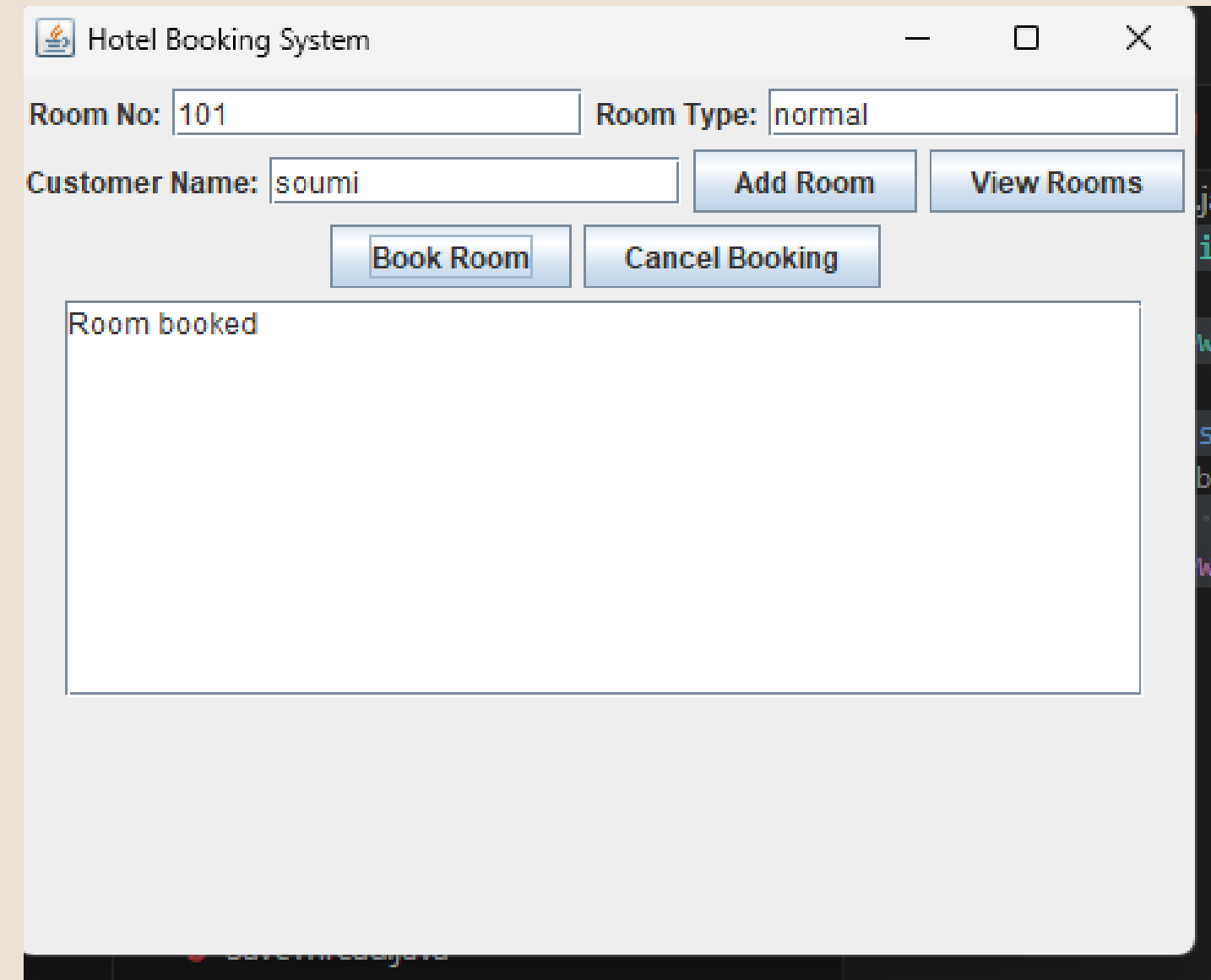
- Booking triggers a thread
- Thread saves data to file

Why Chosen:

- Improves performance
- Prevents UI freezing
- Simulates real-world concurrent systems

Live Demo

- **Show Working System:**
 - Open GUI
 - Add room
 - View rooms
- **Demonstrate Scenario:**
 - Book a room
 - Show booking success
 - Show data saved
- **Error Handling Case:**
 - Try booking same room again
 - Show "Room already booked"



Thank you for listening!